

Elsayed Mustafa

Product AI and Automation Engineer

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SUMMARY

AI & Automation Engineer building LLM agents, backend services and data pipelines that run unattended in production — Python, FastAPI, Supabase, and the integrations between them.

EXPERIENCE

AI & Automation Engineer, BlueBug

02/2026 — Present | Remote

- Build and maintain three internal AI and automation products in Python, FastAPI and Supabase
- Designed resumable scraping workflows with captcha/TOTP handling, account rotation and retry logic
- Shipped every trading strategy behind paper-trading validation and two independent safety switches

AI Automation Developer, Upwork (freelance)

06/2024 — Present | Remote

- Delivered **6+** client platforms end to end, from first call to launch; earned Upwork Rising Talent
- Worked directly with non-technical clients to turn vague requirements into shipped systems
- Delivered a wholesale ops platform with multi-format supplier ingestion, GTIN validation and transaction-safe workflows, covered by **162** automated tests
- Rebuilt a bilingual field-service CRM in Bahrain: migrated **17,000+** records into **17** secured tables and cut the slowest page from **5.6s** to **0.56s**
- Built a resumable lead pipeline covering **2,662** towns with classification, enrichment, deduplication and QA reporting
- Built inventory and marketplace-pricing platforms across Amazon, eBay and Shopify covering **25,000+** items

PROJECTS

Apply Agent

Python, Supabase, Gemini, GitHub Actions, Telegram, Deno

- Built a job-search system pulling from **17** public ATS and aggregator sources, scoring each role against a structured CV with an LLM
- Designed CV tailoring as constrained selection over a closed set of bullet ids, making fabricated experience structurally impossible rather than merely discouraged
- Secured a public webhook with a shared secret, chat allowlisting and strict payload validation, verified against the deployed endpoint
- Covered the system with **258** offline tests including recorded-fixture contract tests that fail when an upstream API changes shape

Amazon ASIN & Review Intelligence Platform

Python, Playwright, Supabase, Google Sheets APIs

- Built six scraping and auditing tools for ASIN details, review collection, variation analysis, listing health and gap detection across marketplaces
- Synchronised reporting across Supabase and Google Sheets so operators could work in the tool they already used

SKILLS

Backend & APIs Python, FastAPI, REST APIs, Webhooks, Authentication, PostgreSQL, Supabase, SQL, SQLite

AI & LLM LLM APIs, RAG, AI agents, Prompt engineering, Structured output, Ollama, Whisper, LightGBM, scikit-learn, Model evaluation, NLP

Automation & Scraping Playwright, Selenium, BeautifulSoup, Google Apps Script, Resumable workflows, Data pipelines, Amazon SP-API, eBay API

Data & Platforms Pandas, NumPy, ETL, Docker, Git, Linux, Streamlit, Google Workspace APIs, Telegram Bots

EDUCATION

B.Sc. Computer and Information Science, Ain Shams University

2020 — 2024

Scientific Computing · Graduation project: data-driven English practice app — conversation simulation and grammar correction

CERTIFICATES

Machine Learning Specialization — DeepLearning.AI · Natural Language Processing with Classification and Vector Spaces — DeepLearning.AI · Advanced SQL Querying — Maven Analytics · Associate Data Analyst in SQL — DataCamp

LANGUAGES

Arabic (native) · English (professional)